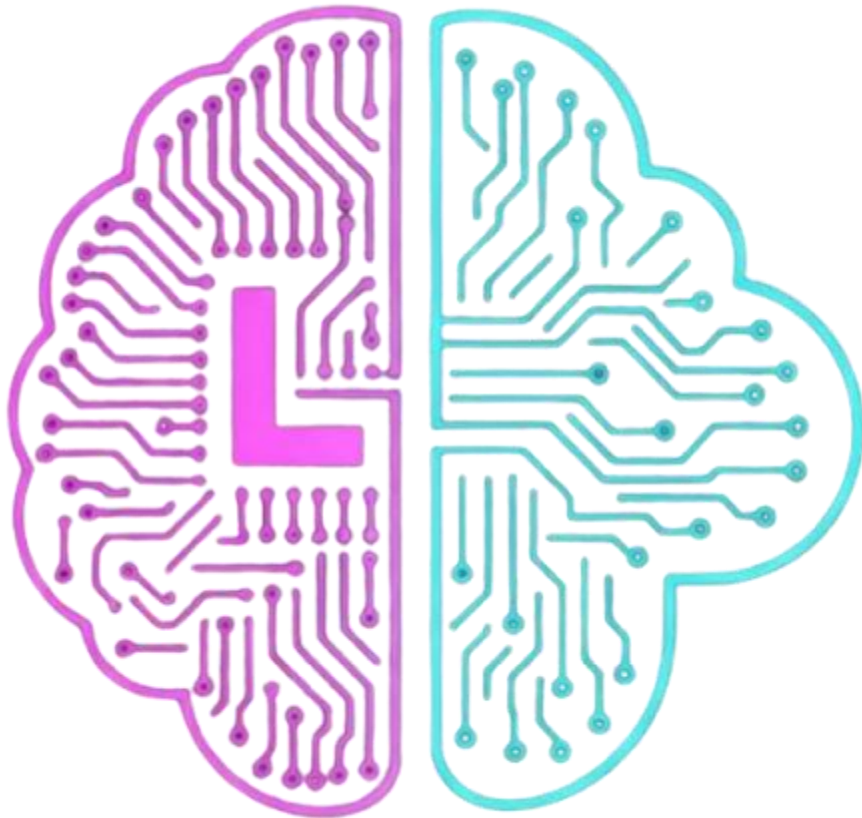




LUMEN AI

Illuminate your learning

TITLE OF THE PAGE

Title of the Project: LUMEN AI- Illuminate your learning

Name of the candidates: Shri. Chetan Mahantesh Badiger
Shri. Mohammed Faaris Khan
Shri. Gaurang Bhushan Gaude
Shri. Ajay Prakash Naik
Shri. Gauresh Gajanan Gaude

Course: Bachelor Of Computer Application
(B.C.A)

Academic Year: 2025-2026

Name of the project guide: Mrs.Diksha Prabhu Khorjuvenkar

Name of the college: GVMS Gopal Govind Poy Raiturcar
College of Commerce & Economics,
Ponda-Goa

DECLARATION

We, the undersigned, hereby declare that the project work entitled Lumen AI: Illuminate your learning has been prepared by us under the guidance of Assistant professor Mrs. Diksha Prabhu Khorjuvenkar. This project is a work submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Computer Applications (B.C.A). The project work has been prepared by us and to the best of our knowledge, it has not previously formed the basis for the award of any Diploma or Degree.

Admission No.	Name of the candidates	Signature
2305002	Shri. Chetan Mahantesh Badiger	
2305004	Shri. Mohammed Faaris Khan	
2305012	Shri. Gaurang Bhushan Gaude	
2305021	Shri. Ajay Prakash Naik	
2305023	Shri. Gauresh Gajanan Gaude	

CERTIFICATE BY GUIDE

This is to certify that the project report/document is a record of work done by the candidates under my guidance during the period of study. To the best of my knowledge, the study carried out and the result obtained, have not been submitted to this or any other university or institute for the award of any degree or diploma.

Asst. Professor in computer science
Mrs.Diksha Prabhu Khorjuvenkar

GOA VIDHYAPRASARAK MANDAL'S
GOPAL GOVIND POY RAITURCAR COLLEGE OF COMMERCE &
ECONOMICS
FARMAGUDI - PONDA GOA



Certified that the project work entitled
“Lumen AI - Illuminate your learning”

Is a bonafide work carried out by

Shri. Chetan Mahantesh Badiger

Shri. Mohammed Faaris Khan

Shri. Gaurang Bhushan Gaude

Shri. Ajay Prakash Naik

Shri. Gauresh Gajanan Gaude

In partial fulfillment for the award of the degree of Bachelor of Computer Applications (B.C.A) of Goa University for the academic year 2024-2025. It is certified that all corrections and suggestions for integrated assessment have been incorporated in the report deposited in the departmental library. The report has been approved as it satisfies the academic requirements of the project work for the Bachelor of Computer Applications.

Internal Guide

Head of the Department

Principal

External Examiner

ACKNOWLEDGEMENT

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ANALYSIS

CHAPTER 1

1. Analysis

With the increasing demand for intelligent and efficient learning tools, AI-powered educational applications are becoming essential in modern academic environments. This project aims to design and develop an educational support application titled Lumen AI: Illuminate Your Learning. The application is specifically designed for students and focuses on automating the note-taking process by allowing users to upload documents or record lecture audio, which is then securely stored and organized into structured data. Lumen AI integrates an AI chatbot that enables students to retrieve information and receive instant answers whenever needed. Additionally, the platform enhances learning through automated flashcards and quizzes, promoting better retention and long-term understanding. Available on Android devices, Lumen AI provides a professional, innovative solution that simplifies information management while supporting students in achieving academic success.

1.1 Proposed System

Artificial intelligence is transforming the way students manage and interact with academic content. Lumen AI is a comprehensive educational support application designed to automate and enhance the note-taking process. The platform allows students to log in, add their classes, select specific lecture dates, and upload audio recordings or documents from their sessions. The system then generates accurate transcripts along with fully summarized and structured notes, enabling students to clearly understand complex topics without spending hours manually organizing information.

The application includes intelligent features such as an AI-powered chatbot that can answer questions based strictly on the recorded class content. Students can select a specific lecture date and request detailed explanations, clarifications, or key points instantly. In addition, Lumen AI automatically generates flashcards and quizzes to test understanding and support exam preparation, reinforcing knowledge retention through active recall and structured practice.

Overall, Lumen AI enhances educational experiences by providing a streamlined and user-friendly platform for academic organization and revision. With reliable long-term storage powered by Supabase and an intuitive Android-based interface, the application prioritizes ease of use and efficiency. By automating note-taking and content structuring, Lumen AI

allows students to dedicate more time to deeper understanding and critical thinking, ultimately transforming how they approach learning in the digital age.

1.1.1 Objectives

The Lumen AI: Illuminate Your Learning application is designed with several key objectives in mind, each focused on improving academic efficiency and reducing the pressures students face in modern educational environments:

1. Enhance Academic Efficiency and Reduce Stress:

Our primary goal is to minimize the time students spend on manual note-taking by automating lecture transcription and summarization. By streamlining this process, Lumen AI allows students to focus on deeper understanding of subject material while reducing academic stress and improving overall exam performance.

2. Automate and Structure Learning Materials:

The application aims to transform unstructured lecture audio and documents into organized, structured notes. By enabling students to upload recordings or files based on specific classes and dates, Lumen AI ensures that information is clearly categorized, easily accessible, and academically meaningful.

3. Provide Instantaneous Academic Support:

An AI-powered chatbot is integrated within the platform to function as a virtual academic assistant. This feature allows students to ask detailed questions related to specific lectures and receive accurate responses based solely on their recorded content, fostering an efficient and responsive learning environment.

4. Strengthen Exam Preparation and Knowledge Retention:

The application automatically generates flashcards and quizzes designed to test understanding and reinforce key concepts. This objective supports active recall practices, helping students prepare more effectively for assessments and long-term knowledge retention.

5. Support Diverse Educational Levels:

Lumen AI is designed to accommodate learners ranging from high school students to university and even PhD candidates. By providing structured academic assistance across various subjects and complexity levels, the platform promotes independent and adaptable learning.

6. Expand Accessibility and Future Integration:

Currently developed as a prototype and offered free of charge, Lumen AI aims to expand its accessibility by launching on major app stores and developing a web-based platform for laptop users. Future integration with platforms such as Google Classroom is envisioned to enhance usability and ensure seamless adoption within existing educational ecosystems.

1.1.2 Features

The Lumen AI application introduces a focused and practical set of core features designed to streamline academic workflows and enhance productivity for students. Each functionality has been carefully developed to automate essential study tasks while maintaining simplicity and efficiency within the Android platform. Below is a detailed overview of its primary features and capabilities.

1. Automated Lecture Processing and Smart Note Generation

Lumen AI supports multiple file formats, including documents and audio files such as PDF, DOCX, and MP3. After a user uploads lecture material and selects the appropriate class and date, the system processes the content and generates a raw results page containing both a full transcript (for audio recordings) and an AI-generated summary. This automation significantly reduces the time students spend manually writing notes. While summaries are initially structured around main topics, students can further refine or reorganize content through interaction with the chatbot, enabling clearer academic organization.

2. Context-Based AI Chatbot Assistance

The application integrates an AI chatbot designed to answer questions strictly based on the uploaded and processed lecture materials. Students can select specific class dates and request clarification on recorded content, ensuring responses remain relevant and context-specific. Although the chatbot currently focuses on direct question-and-answer functionality without advanced adaptive learning features, it serves as an immediate academic reference tool, reducing the need to manually search through notes.

3. Automated Flashcards and Multiple-Choice Quiz System

To strengthen exam preparation and knowledge retention, Lumen AI automatically generates flashcards and multiple-choice quizzes derived from processed lecture content. Flashcards are system-generated and designed to reinforce key concepts through active recall techniques. The quiz feature provides multiple-choice questions and displays scores upon completion, enabling students to assess their understanding before examinations. While flashcards are not currently editable, the automated system ensures consistency and structured revision support.

4. Academic Organization and Class Management System

Lumen AI allows users to create and manage their own subjects or classes based on individual academic needs. Students can assign uploaded materials to specific classes and dates, helping maintain chronological organization. Although filtering options are currently limited to calendar-based navigation and class name search functionality, the structure provides a foundational academic management system that centralizes lecture materials within one accessible platform.

5. User Interface and System Infrastructure

Developed as an Android prototype, Lumen AI offers a partially minimal interface with dark mode support to enhance usability and visual comfort. The application operates

without offline access and currently utilizes Supabase under its free-tier limitations for backend services. While cloud synchronization and advanced storage optimization are not yet implemented, the infrastructure provides a functional foundation for future scalability and deployment to major app distribution platforms.

1.1.3 Intended Audience

The Lumen AI application has been strategically developed to support a wide range of learners by addressing common academic challenges such as time management, information overload, and exam preparation. The platform is designed to assist students across multiple educational levels by automating note-taking and organizing lecture content into structured, accessible formats. Below is a detailed overview of the intended audience and how the application serves their specific academic needs.

1. High School Students

One of the primary target groups for Lumen AI is high school students, who often face increasing academic pressure and dense curricular content.

Functionality:

By allowing students to upload lecture recordings or documents, the application automatically generates transcripts and concise summaries, reducing the burden of manual note-taking. The integrated chatbot enables learners to ask questions about specific lessons, while auto-generated flashcards and multiple-choice quizzes support revision and exam preparation. This structured and automated approach helps high school students better manage their workload, improve comprehension, and reduce academic stress.

2. University Students

University learners represent another significant audience, particularly due to the volume and complexity of lecture-based learning.

Functionality:

Students can create and organize multiple classes, assign lectures by date, and retrieve summarized content instantly. This is especially beneficial for large lecture environments where note-taking can be overwhelming. The ability to generate quizzes and review flashcards directly from lecture material enhances retention and supports efficient preparation for midterms and final examinations.

3. Postgraduate and Research Students

Lumen AI is also suitable for postgraduate and PhD students who manage advanced theoretical material and extensive academic discussions.

Functionality:

The automated transcription and summarization system assists in documenting detailed lectures, seminars, or research discussions. By centralizing content within structured class categories, the platform simplifies academic organization, enabling advanced learners to

focus more on critical analysis and research development rather than administrative note management.

1.2 Hardware and Software Specification

1.2.1 Hardware Requirements

1. Development Environment:

The development of Lumen AI is conducted on systems running the Windows operating system, utilizing VS code as the primary development environment. VS code provides the necessary tools for coding, testing, debugging, and performance optimization to ensure stable application deployment on supported Android devices.

2. User Device Compatibility:

Lumen AI is designed exclusively for Android devices running Android version 8.0 or higher. The application supports both smartphones and tablets to maximize accessibility across different device types.

Device Requirements:

- Minimum 4GB RAM for stable performance
- Minimum available storage space below 2GB for installation and operational data
- Functional microphone access (user permission required) for recording lecture audio
- Stable internet connectivity for AI processing and data handling

1.2.2 Software Requirements

Flutter



Flutter is an open-source UI software development toolkit developed by Google that enables developers to build natively compiled applications from a single codebase. It is widely recognized for its ability to create high-performance, visually consistent, and responsive mobile applications. Flutter uses a rich set of pre-designed widgets that follow modern design principles, allowing developers to construct intuitive and aesthetically pleasing user interfaces efficiently. Its hot reload feature significantly accelerates development by enabling real-time updates without restarting the application.

In the Lumen AI application, Flutter is responsible for constructing the entire Android user interface, including authentication screens, class management pages, lecture upload interfaces, chatbot interaction windows, and quiz modules. Its reactive framework ensures smooth transitions between screens and efficient rendering of dynamic content such as transcripts and summaries. Additionally, Flutter's scalable architecture makes it suitable for potential future expansion to other platforms, including web and iOS, without requiring a complete rewrite of the codebase.

Dart



Dart is a modern, object-oriented programming language developed by Google and serves as the primary language for Flutter development. It is optimized for building fast and scalable applications, supporting both just-in-time (JIT) compilation for development efficiency and ahead-of-time (AOT) compilation for optimized production performance. Dart emphasizes readability, structured programming, and strong typing, making it suitable for maintaining clean and maintainable codebases.

Within Lumen AI, Dart handles the application's internal logic, state management, and user interactions. It manages processes such as user authentication, lecture uploads, date selection, API communication, and display of processed results. Dart also supports asynchronous programming, which is essential for handling file uploads and waiting for

AI-generated transcripts and summaries without freezing the interface. This ensures a smooth and responsive experience for users while interacting with AI-driven features.

FastAPI



FastAPI is a high-performance web framework built with Python for developing APIs quickly and efficiently. It is designed to create secure and scalable backend systems with automatic request validation and interactive documentation support. FastAPI is particularly valued for its speed, as it is built on modern asynchronous programming standards that allow handling multiple requests concurrently.

In Lumen AI, FastAPI functions as the backbone of backend communication. It manages incoming requests from the Flutter application, processes uploaded files, and coordinates interactions with external AI services such as the Gemini API. FastAPI ensures structured data exchange between the frontend and backend while maintaining performance efficiency. Its ability to handle asynchronous operations is especially important for

managing AI processing tasks, which may require several seconds to generate transcripts and summaries.

Python



Python is a versatile and widely adopted programming language known for its simplicity, readability, and extensive ecosystem of libraries. It is extensively used in artificial intelligence, machine learning, backend development, and data processing due to its flexibility and strong community support. Python's clean syntax enables efficient development and easier maintenance of complex systems.

In the Lumen AI architecture, Python is utilized alongside FastAPI to manage backend operations and AI-related processing. It facilitates communication with the Gemini API for transcription, summarization, and chatbot responses. Python handles data processing workflows, ensuring that uploaded lecture materials are structured properly before being

returned to the mobile application. Its adaptability allows seamless integration of future AI enhancements or additional backend features as the application evolves.

Supabase

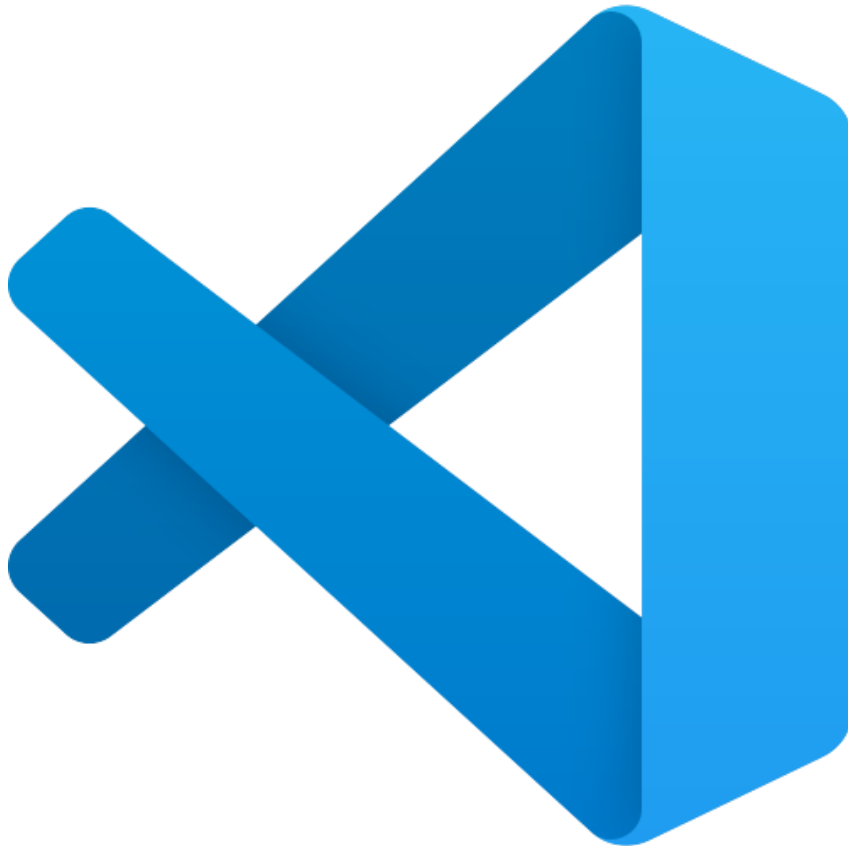


Supabase is an open-source backend-as-a-service platform that provides database management, authentication services, and storage solutions. Built on PostgreSQL, Supabase offers structured relational database capabilities combined with modern backend features. It simplifies backend development by eliminating the need to manually configure complex server infrastructure.

For Lumen AI, Supabase is responsible for managing user authentication and storing structured academic data. It maintains user accounts, class records, lecture metadata, transcripts, summaries, flashcards, and quiz results within its database system. Operating under the free-tier plan, Supabase provides reliable backend support while maintaining

cost efficiency during the prototype phase. Although limited by storage capacity, it establishes a scalable foundation for future system expansion and improved data management.

Visual Studio Code (VS Code)



Visual Studio Code (VS Code) is a lightweight yet powerful source-code editor developed by Microsoft. It supports a wide range of programming languages and offers extensive extension capabilities that enhance productivity, debugging, and version control integration. VS Code is widely preferred by developers due to its customizable environment and efficient performance.

During the development of Lumen AI, VS Code is used as a primary development tool for writing and managing backend and application code. It supports Dart and Flutter extensions for mobile development, as well as Python and FastAPI configurations for backend implementation. Integrated debugging tools, syntax highlighting, and Git version control features streamline collaboration and code maintenance. VS Code's flexibility

ensures an organized and efficient development workflow, contributing to the structured implementation of Lumen AI's features.

1.4 Gantt Chart

A Gantt chart is a planning tool used in project management to organize and monitor a project's timeline. It clearly displays when each task begins and ends, how tasks are connected to one another, and the progress made on each activity. By presenting this information in a visual format, it allows project managers to see the overall status of the project at a glance. Gantt charts are widely used across different industries to allocate resources effectively, keep track of milestones, and help ensure projects are finished on schedule.

Lumen AI project

Illuminate your learning.

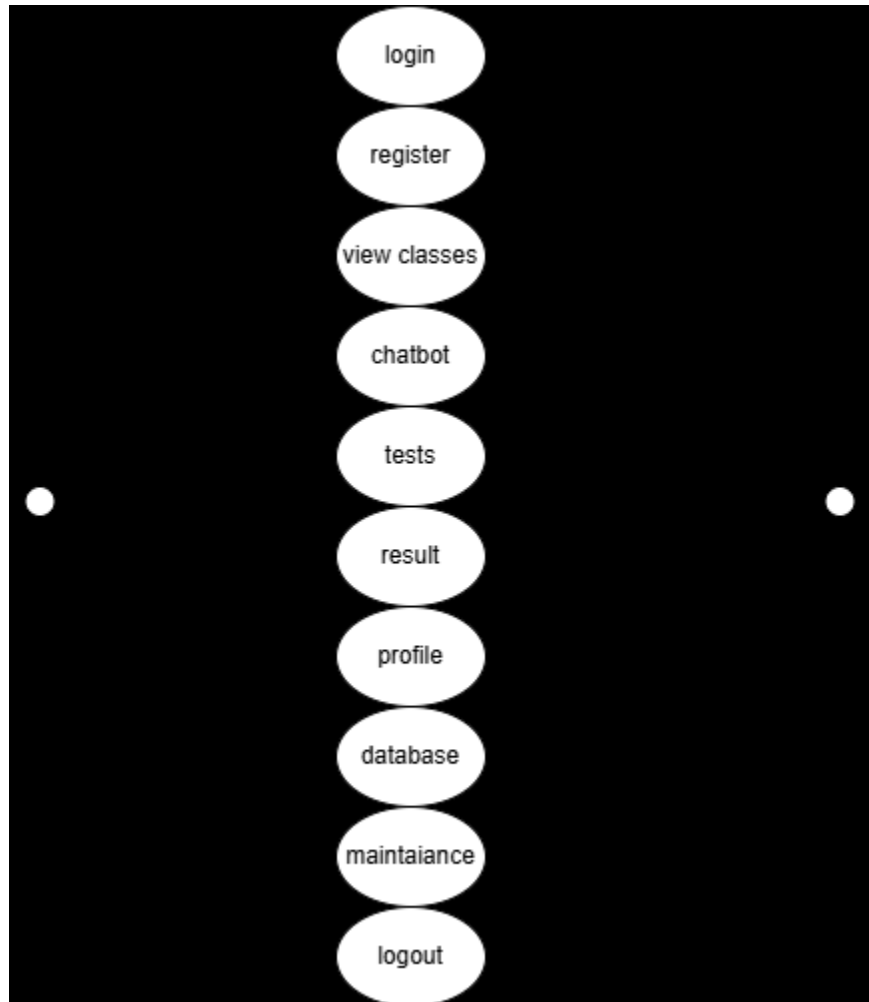
SIMPLE GANTT CHART

TASK	ASSIGNED TO	PROGRESS	START	END
Timeline				
Requirement Identification	team	100%	10-12-2025	11-12-2025
Study of Existing Softwares	team	100%	11-12-2025	12-12-2025
Identification of Features	team	100%	12-12-2025	14-12-2025
Feasibility Study For Project	team	100%	14-12-2025	15-12-2025
Timeline Planning	team	100%	14-12-2025	20-12-2025
Design of ER , Class ,Use case ,Flow Daigrams	team	100%	20-12-2025	01-01-2026
Design Logo	Faaris,Ajay	100%	25-12-2025	28-12-2025
UI Design	Faaris,Gauresh	100%	25-12-2025	02-01-2026
Front-End	Faaris,Ajay,Gauresh	100%	02-01-2026	30-01-2026
Back-End	Faaris,Gaurang,Chetan	100%	10-01-2026	02-02-2026
Database connectivity	Faaris,Gaurang	100%	21-01-2026	02-02-2026
Validation Testing	Chetan,faaris	100%	02-02-2026	10-02-2026
Unit Testing	Chetan,Gaurang	100%	06-02-2026	12-02-2026
System Integration	Chetan,Ajay	100%	06-02-2026	23-02-2026
Integration Testing	Chetan,Gauresh	100%	06-02-2026	23-02-2026
Documentation	team	100%	10-02-2026	26-02-2026

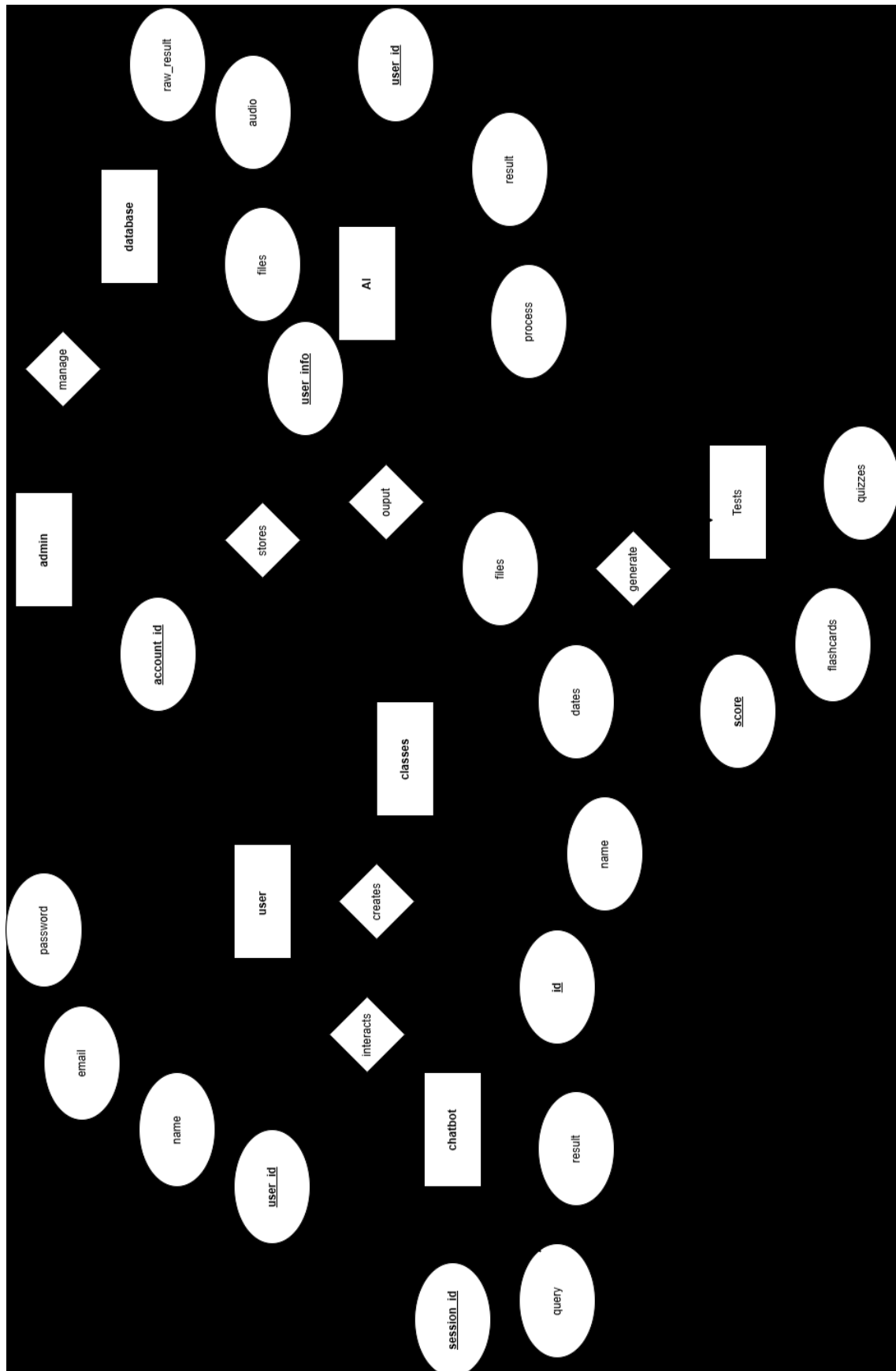
DESIGNS

CHAPTER 2

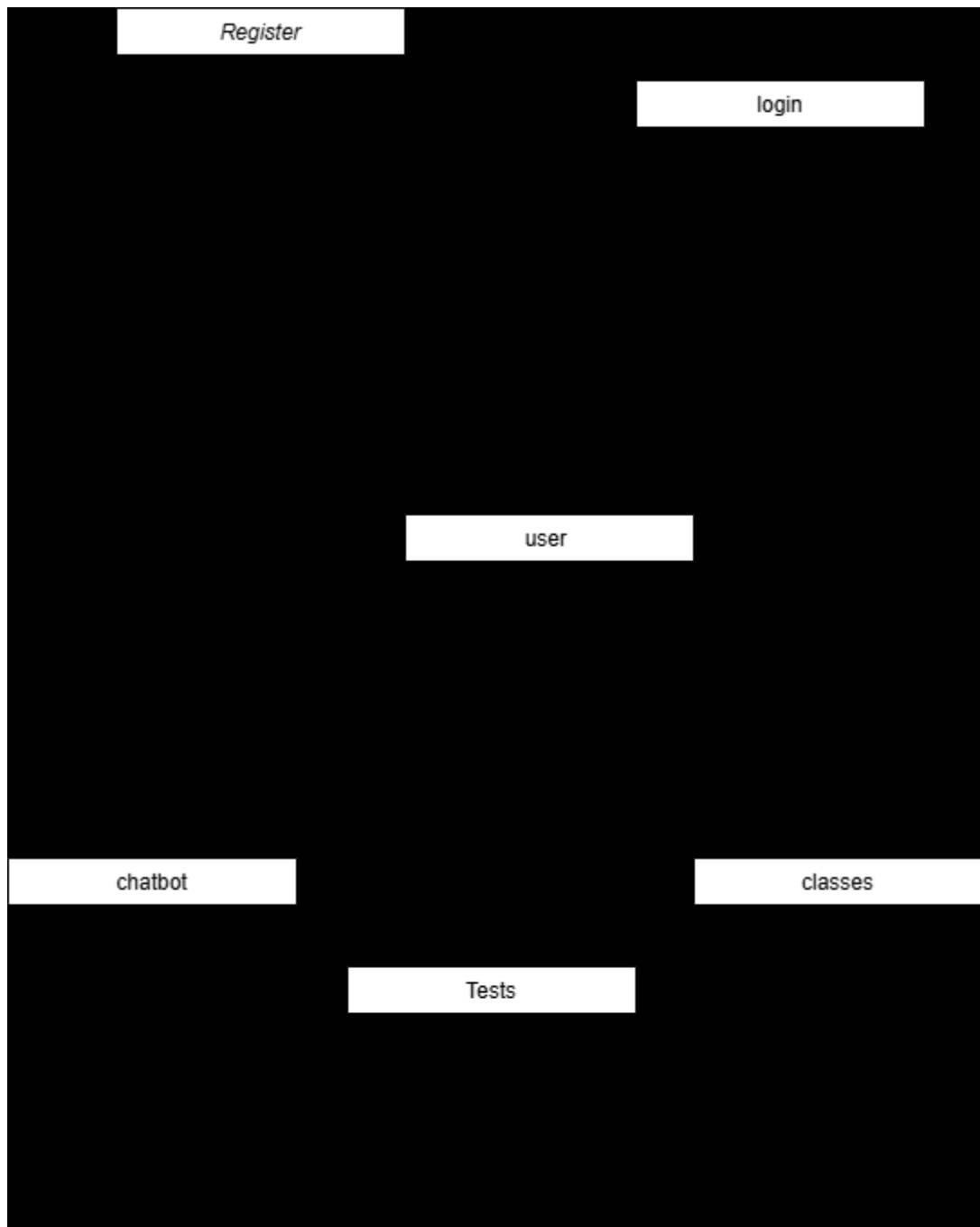
2.1 Use Case Diagram



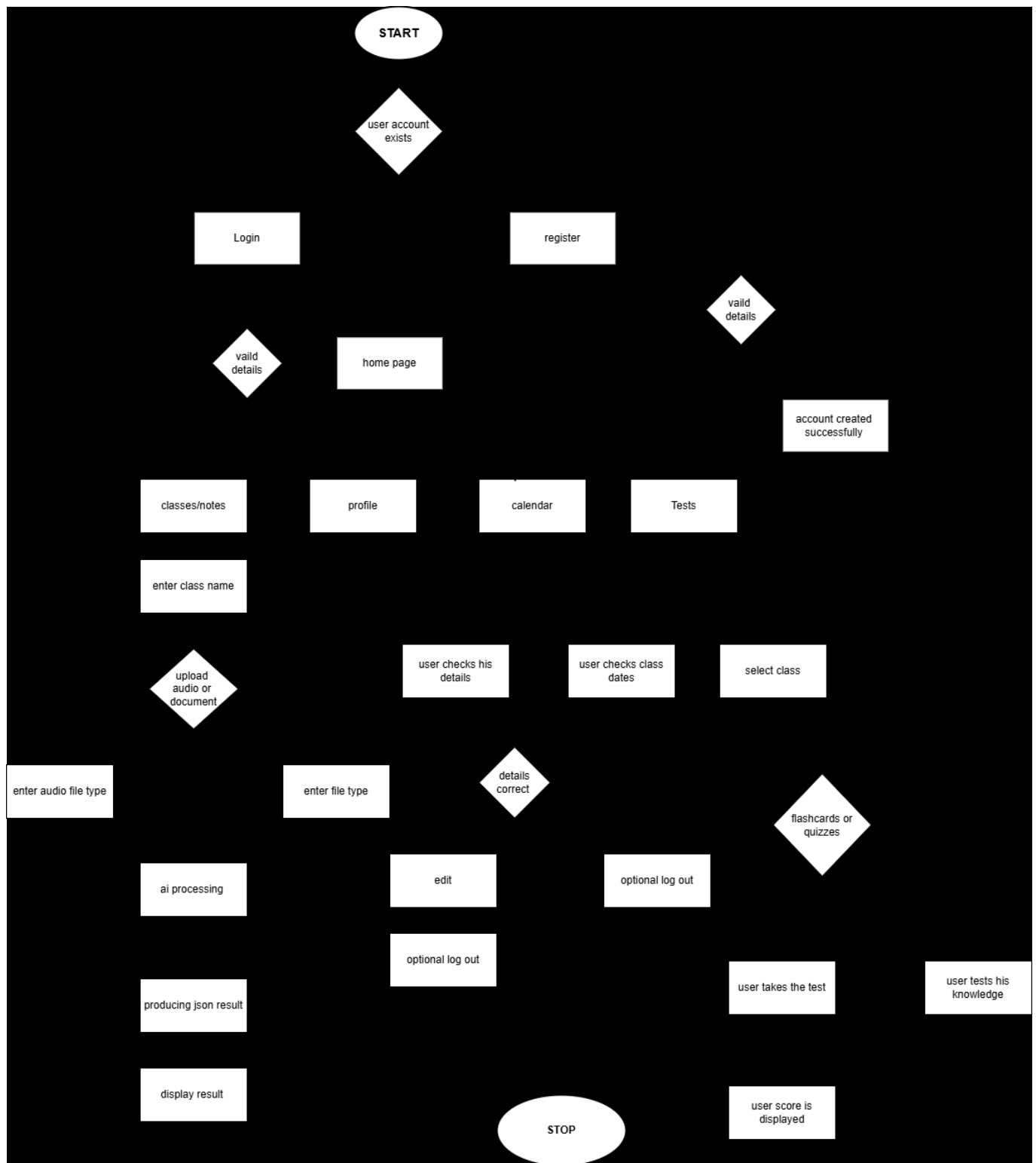
2.2 ER Diagram



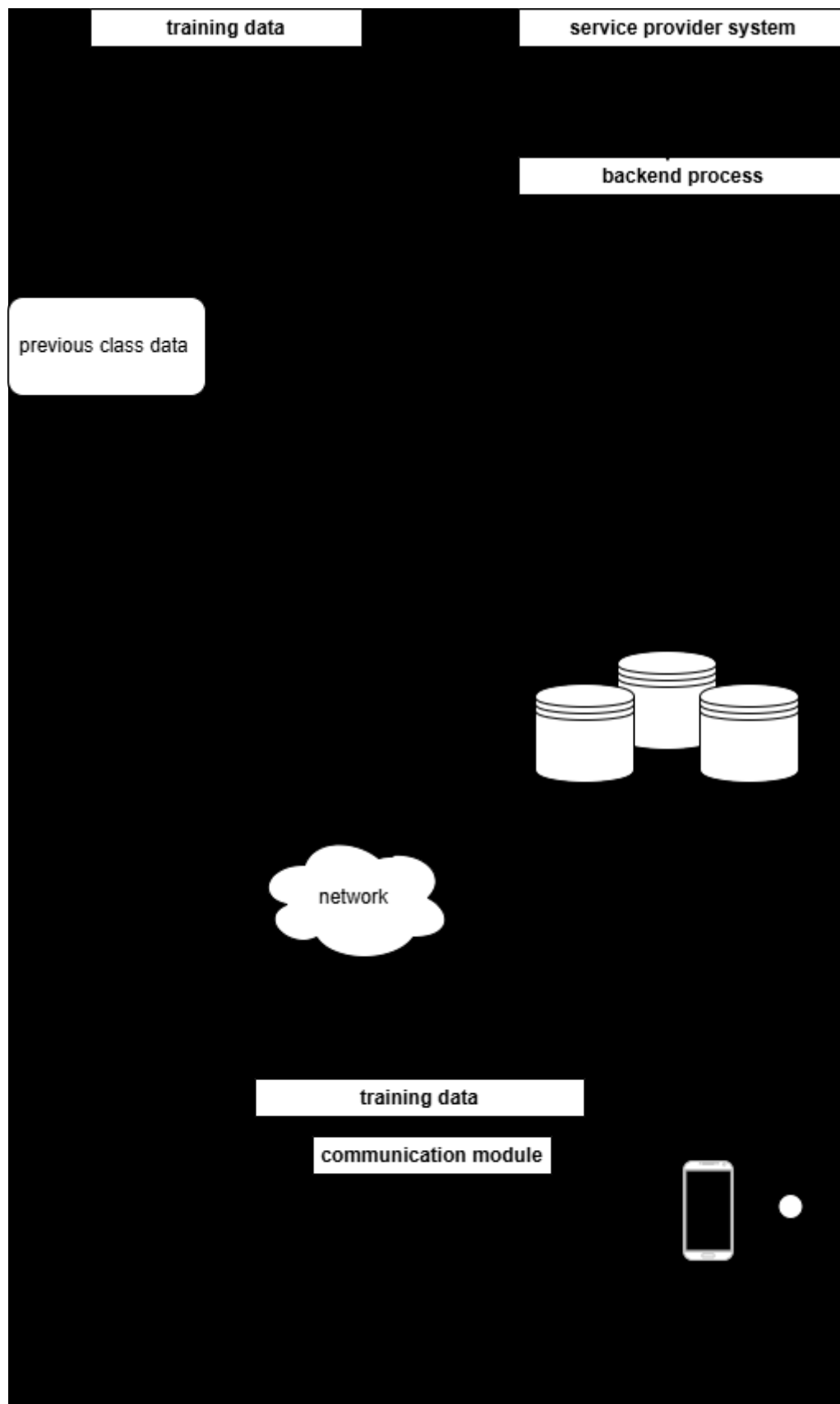
2.3 Class Diagram



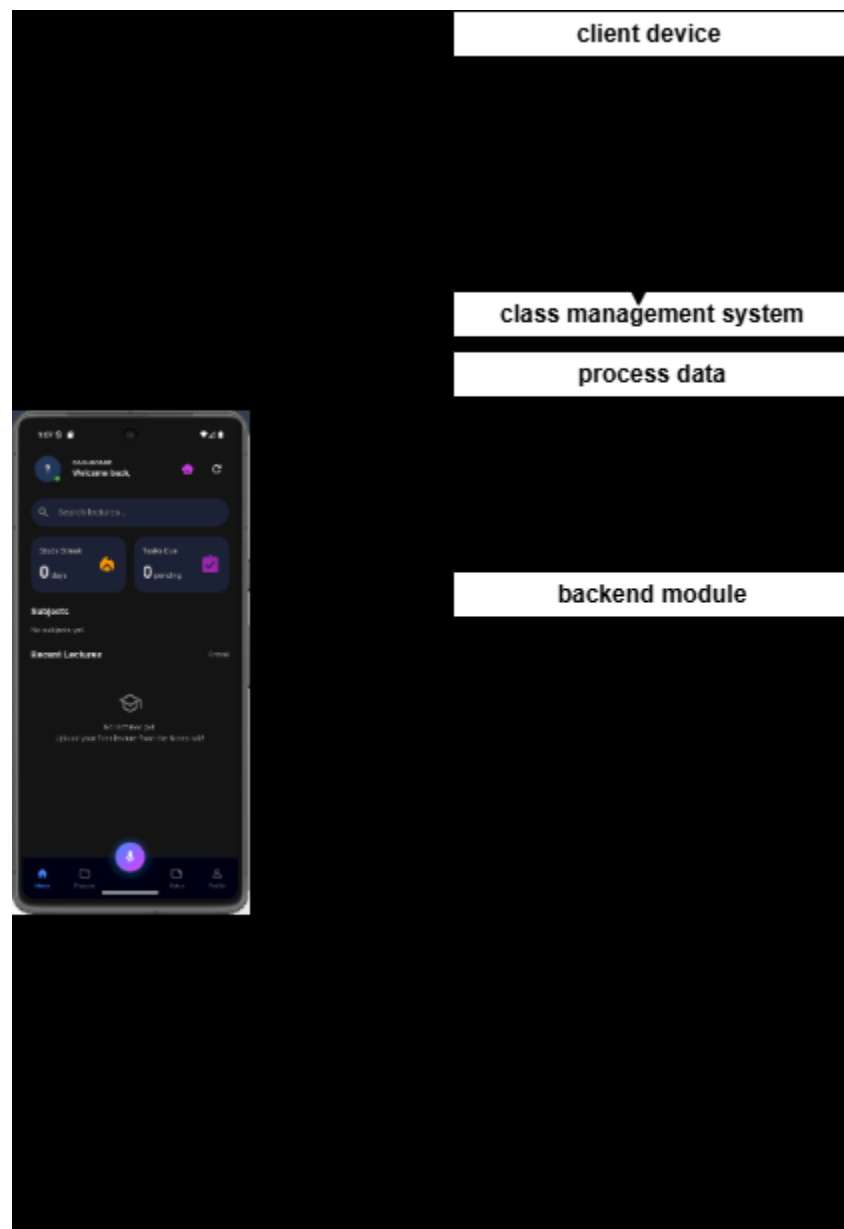
2.4 Flowchart Diagram



2.5 Customer Interaction Diagram



2.6 Client Device Diagram



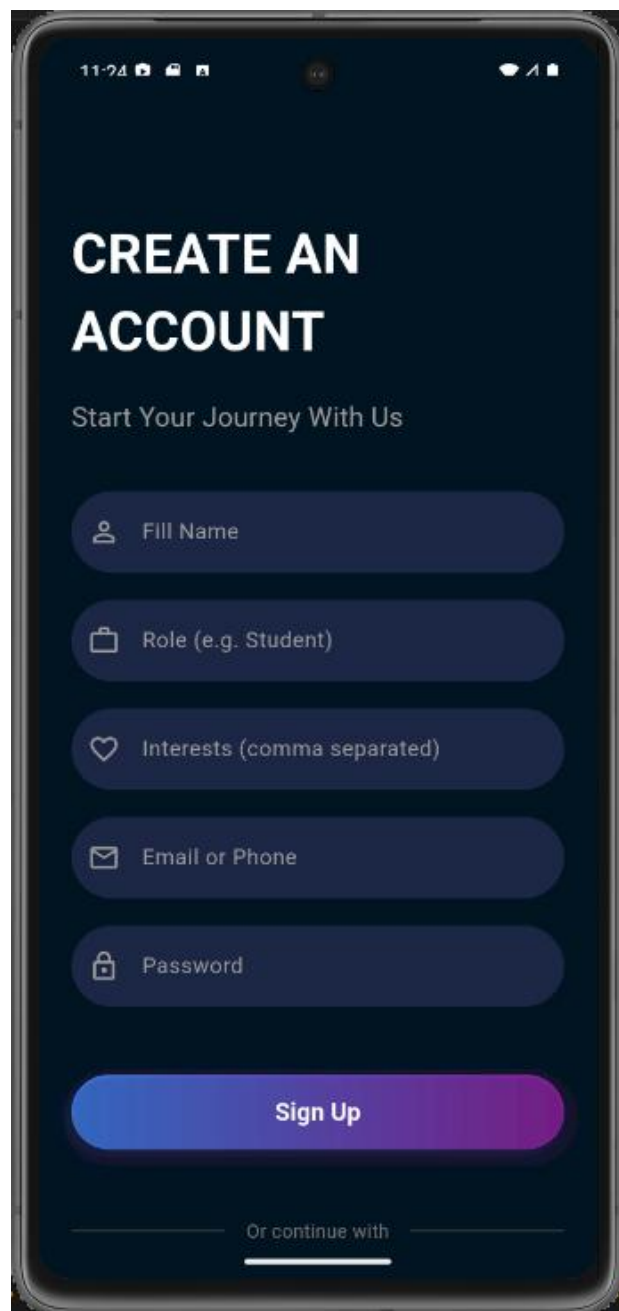
IMPLEMENTAION

Chapter 3

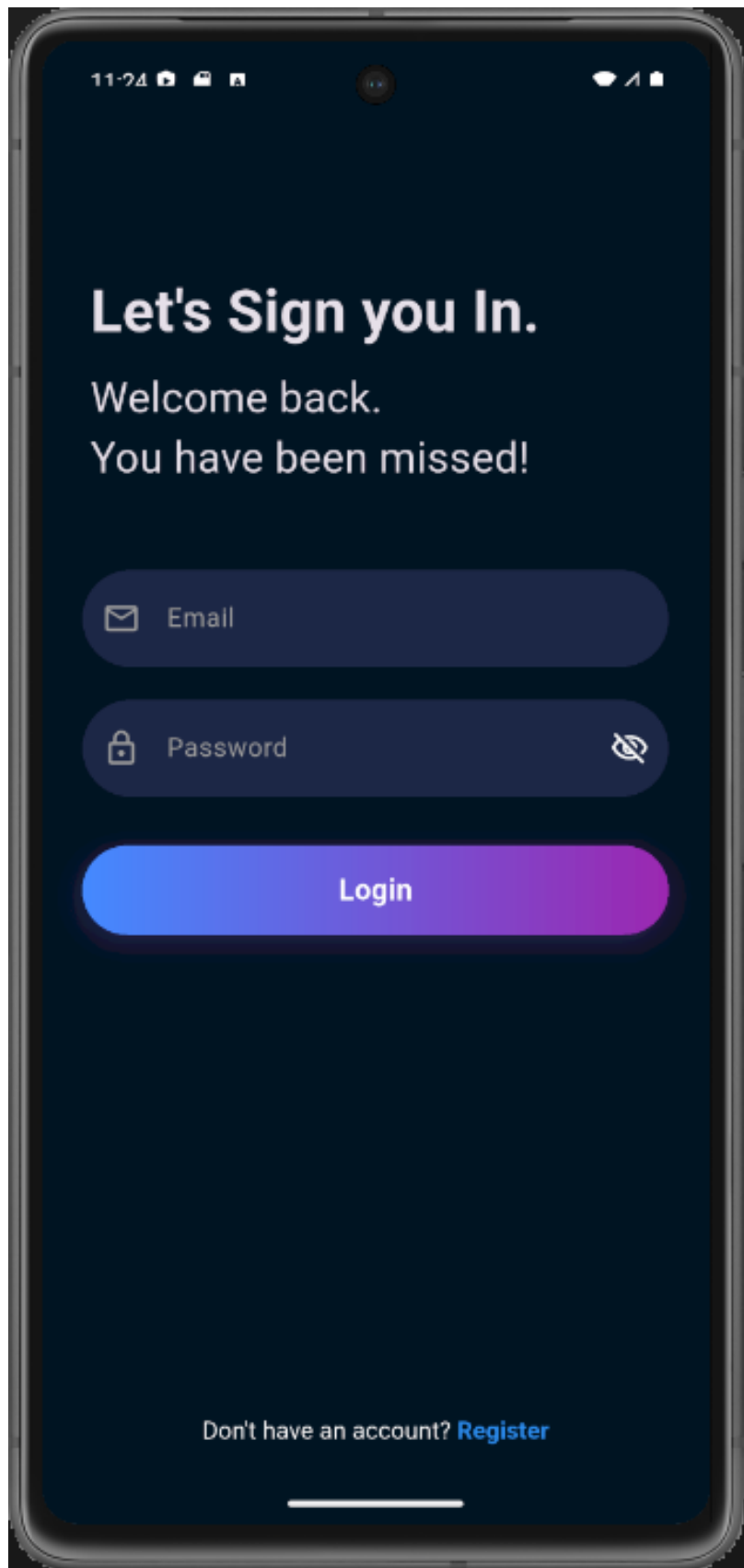
3.1 User Interface

3.1.1 Auth Module

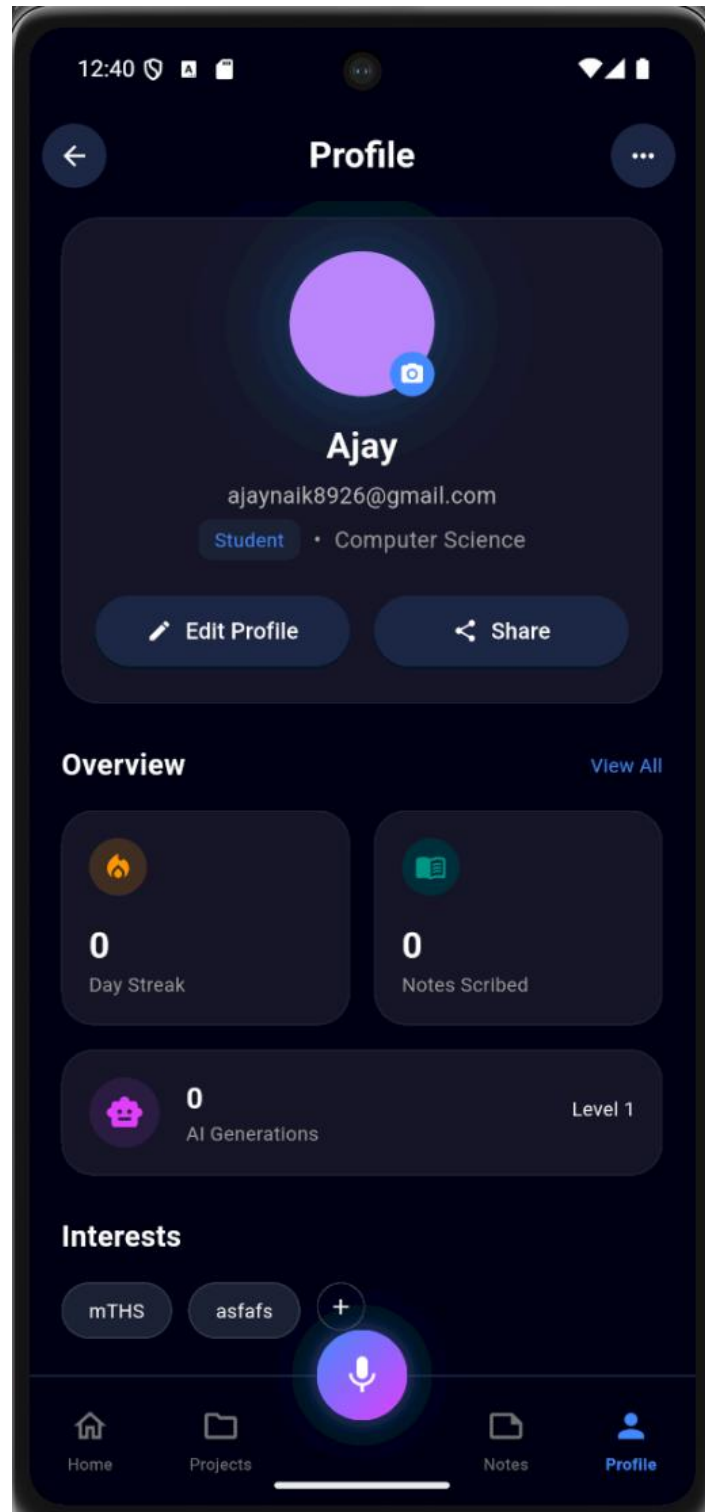
Register page

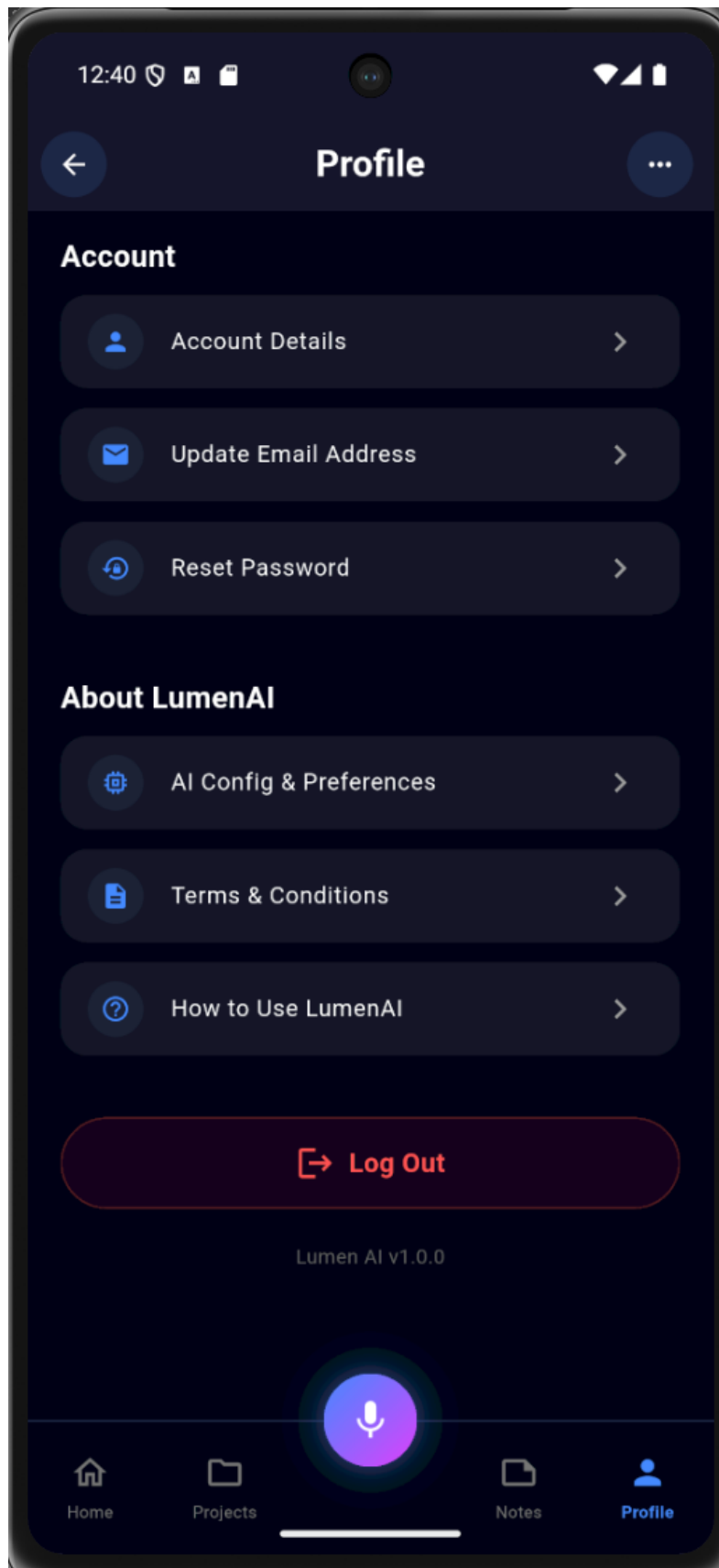


Login page

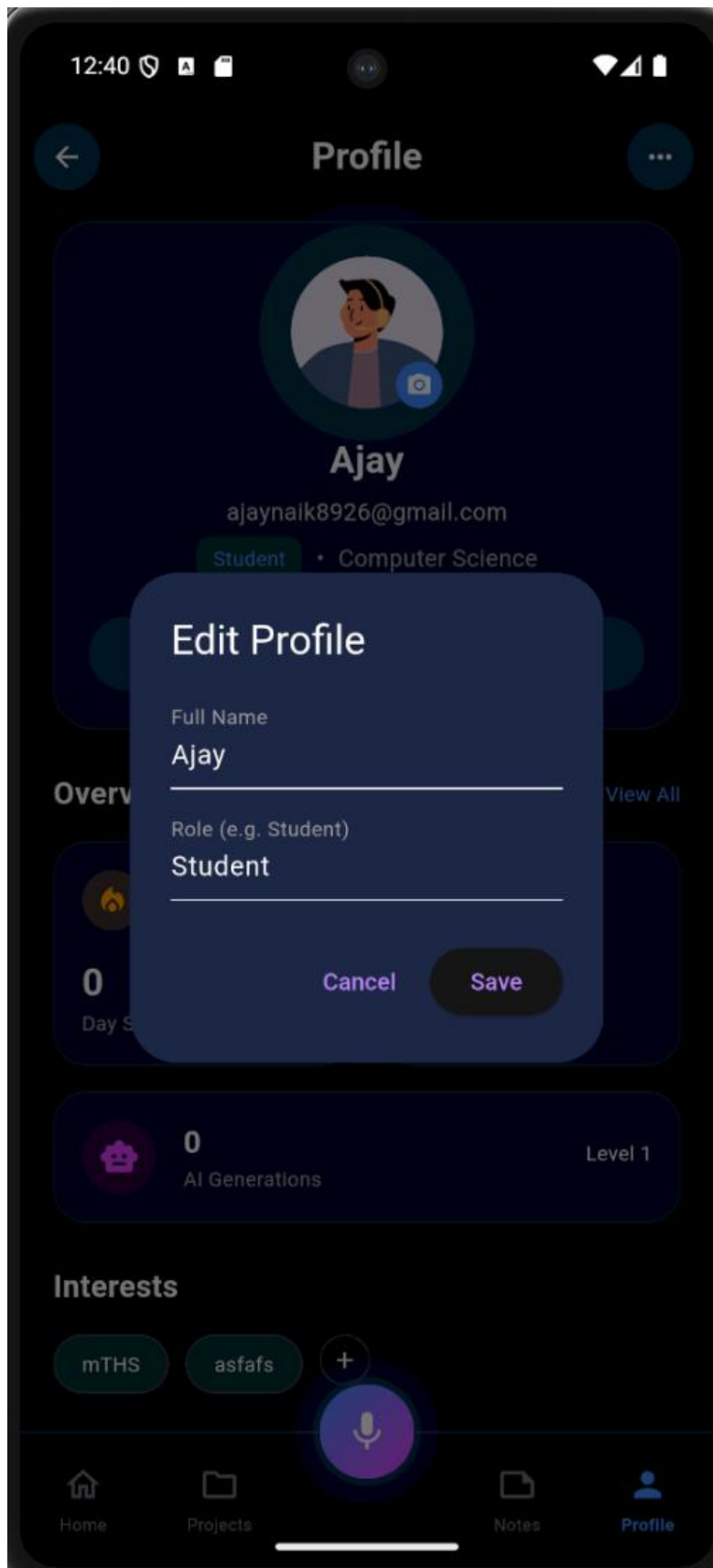


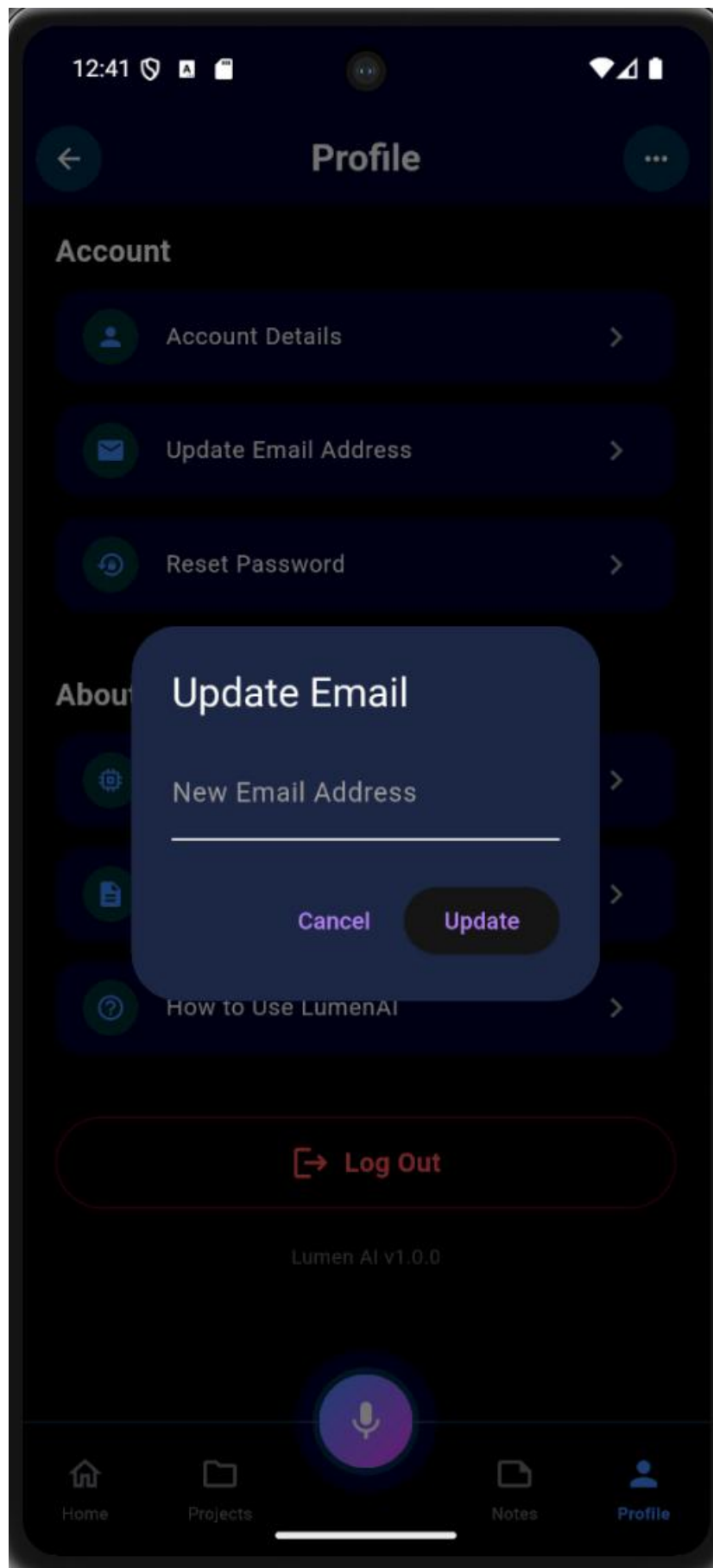
3.1.2 App User Profile



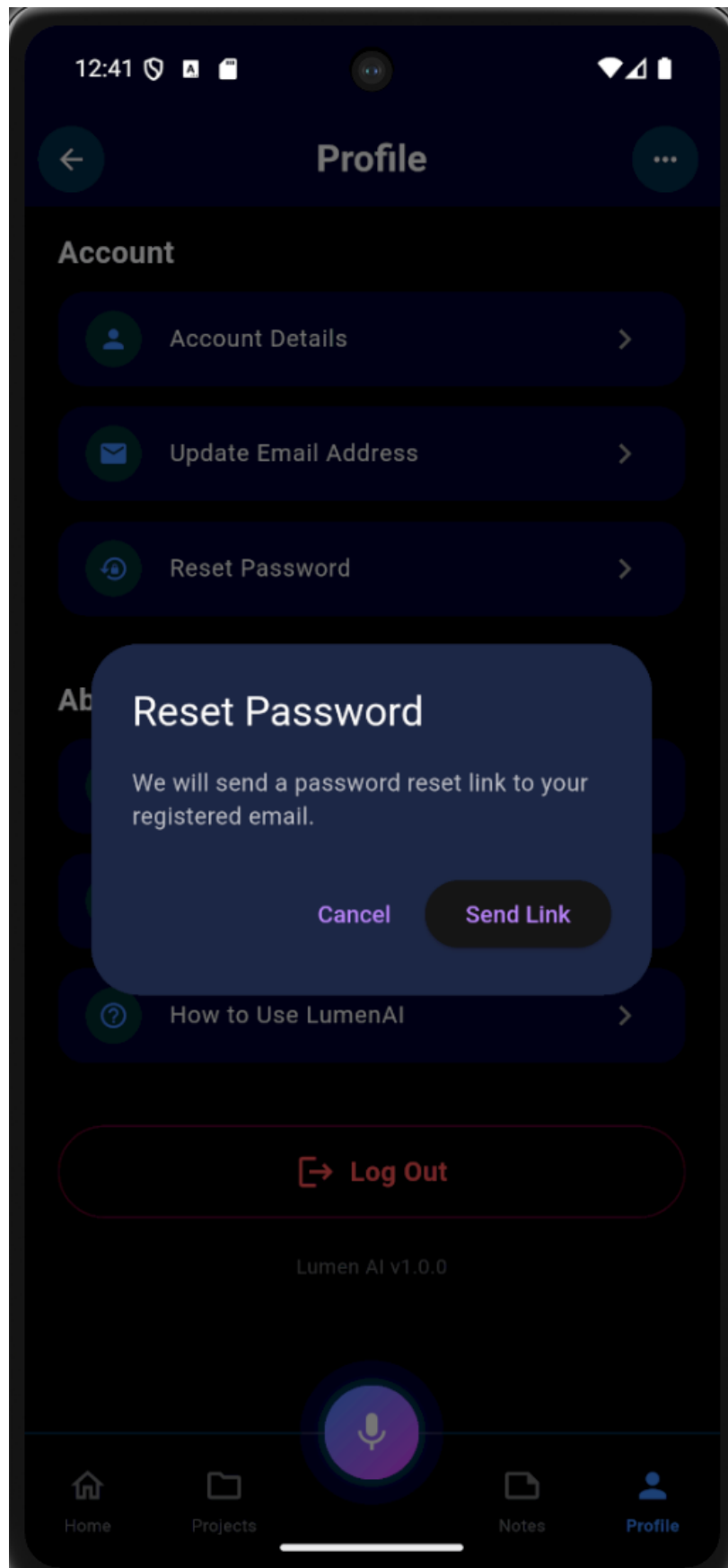


Edit Information





Reset password



3.2 Functionality Design

3.2.1 Notes

3.2.2 Calendar

3.2.3 Chatbot

3.2.4 Flashcards

3.2.5 Quizzes

3.2.6 Results

CONCLUSION AND FUTURE GOALS

Chapter 4

4.1 Conclusion

The development of Lumen AI represents a significant step toward integrating artificial intelligence into everyday academic workflows. This project was designed with a clear objective: to reduce student stress, automate time-consuming note-taking processes, and enhance overall academic performance. In an era where students are constantly managing dense lecture materials, multiple subjects, and limited time, Lumen AI provides a structured and intelligent solution that simplifies learning management.

Throughout the development process, the application was built to function exclusively on Android devices running version 8.0 and above, ensuring compatibility with a wide range of modern smartphones and tablets. By leveraging Flutter and Dart for the frontend, the project achieved a responsive and intuitive user interface capable of handling dynamic content such as transcripts, summaries, flashcards, and quizzes. The integration of FastAPI and Python on the backend enabled efficient communication between the mobile application and AI services, while Supabase provided reliable database management and user authentication support.

A key achievement of this project is the successful implementation of automated lecture processing. Students can upload audio recordings or academic documents in supported formats, after which the system generates full transcripts and concise summaries. This automation eliminates the need for manual note-taking, allowing learners to focus more on understanding concepts rather than transcribing information. The AI-powered chatbot further strengthens this functionality by enabling students to ask questions based strictly on their recorded lectures, ensuring relevant and context-specific responses.

In conclusion, Lumen AI successfully fulfills its primary mission of automating academic organization and reducing cognitive overload for students across high school, university, and postgraduate levels. By combining intelligent automation, structured content management, and AI-driven assistance within a single mobile platform, the project demonstrates how technology can meaningfully enhance modern education.

4.2 Future Goals

As Lumen AI continues to evolve beyond its prototype stage, several strategic future goals have been identified to enhance its functionality, scalability, and overall impact within the educational technology landscape. These planned developments aim to strengthen accessibility, introduce sustainable monetization strategies, improve personalization, and expand platform availability while maintaining the core mission of reducing academic stress and automating learning workflows.

1. Multiple languages and Audio Processing Enhancement

expand compatibility with multiple languages, particularly in audio processing and transcription. Currently, lecture recordings are processed only with English however, future development will focus on integrating multilingual speech recognition systems. This enhancement will allow students from diverse linguistic backgrounds to upload and process lectures delivered in various languages. By incorporating multilingual audio transcription and summarization, Lumen AI can broaden its accessibility to international students and institutions. Such expansion not only increases inclusivity but also strengthens the application's potential for global adoption in academic environments.

2. Monetization Through Premium Subscription Plans

Premium subscription plans as part of a structured monetization model. While Lumen AI currently operates without paid features, introducing tiered premium plans would ensure long-term sustainability and continued technological improvement. Premium offerings may include expanded storage capacity beyond free-tier backend limitations, faster AI processing, advanced summarization options, enhanced chatbot functionality, and priority access to newly released features.

3. Customized Assessment Generation

The current system automatically generates multiple-choice quizzes; however, future iterations will incorporate fully customized test generation. This feature would allow students to select difficulty levels, define specific topics within a lecture, choose question formats such as short-answer or scenario-based questions, and determine the number of questions per test.

4. Deployment to Major App Stores

The official release of Lumen AI on the Google Play Store and Apple App Store will increase visibility, credibility, and user trust. Publishing on these platforms requires compliance with industry standards, performance optimization, and enhanced security measures. Successfully achieving store deployment will not only expand the user base but also position Lumen AI as a competitive educational technology solution within the global marketplace.

5. Audio-Responsive Chatbot Integration

chatbot with audio-listening capability is a forward-looking innovation goal. This enhancement would allow the chatbot to actively process and respond to spoken queries, enabling hands-free interaction. By incorporating speech-to-text and text-to-speech technologies, students could verbally ask questions about their lectures and receive spoken explanations in return. Such functionality would significantly improve accessibility.

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<https://www.geeksforgeeks.org/flutter/flutter-tutorial/>

<https://curriculumredesign.org/our-work/artificial-intelligence-in-education/>